Real-Time versus Reliable: Evaluating the Accuracy and Trustworthiness of VGI Avalanche Reporting

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Abstract

Avalanche forecasting is critical for ensuring backcountry safety, however, traditional forecasting methods are limited by the lack of geographic specificity and infrequent updates(Harvey, 2002; Techel et al., 2015). This essay examines the legitimacy of Volunteered Geographic Information as an additional tool for avalanche hazard assessment for backcountry users by evaluating its reliability and trustworthiness.

Through a comprehensive literature review and the incorporation of studies by Connors et al. (2011), Kocaman and Gokceoglu (2018), Flanagin and Metzger (2008), in addition to other key studies, four key themes were identified: geographic clustering, discrepancies in accuracy, social biases, and variability and subjectivity in risk terminology (Hogeweij, 2019; Mannberg et al., 2020). Accompanying this information and data from Avalanche Canada supports these findings and is strategically included.

Although VGI offers real-time, localized information, its reliability and accuracy can be impacted by inconsistencies and subjective reporting. This study suggests a forecasting model that integrates validated, standardized VGI reports with professional forecasts, to provide a more useful tool for backcountry decision making.

Introduction

Avalanches are an incredibly unpredictable and dangerous hazard for backcountry users, and with the increasing popularity of backcountry recreation, ensuring safety through effective avalanche forecasting has become increasingly important. According to recent avalanche data, approximately 22 people die each year from avalanche related accidents in British Columbia(BC Coroners Service, 2018). Traditional methods such as those provided by Avalanche Canada rely on expert analysis of snowpack conditions, weather patterns and historical records(Harvey, 2002; Techel et al., 2015). While these methods are well established and credible, they have two significant limitations, professional Avalanche forecasts are usually released for a large geographic area lacking geographic specificity, and they are released at 24 or 48 hour intervals at minimum, meaning that they cannot update in real time as conditions change rapidly.

To address these gaps, Volunteered Geographic Information(VGI) avalanche reporting systems have emerged as a potential alternative. VGI refers to geographic data that is voluntarily created and shared by individuals rather than traditional methods, essentially using "citizens as sensors"(Goodchild, 2007). The potential implications of reliable VGI avalanche reporting extend beyond the individual safety decisions and can impact other things in related industries such as search and rescue resource allocation, and park and land management. The challenges identified in this study are also reflective of broader issues within the field of GIScience regarding the integration of VGI data with professional/authoritative sources.

Through the use of user submitted reports of localized snow conditions, weather, and observed avalanche activity, VGI avalanche reports provide a more localized, real time picture of avalanche conditions than professionally generated avalanche forecasts/reports (Connors et al., 2011; Kocaman & Gokceoglu, 2018). However, the use of VGI reports as a potential tool for avalanche risk analysis and the usage of it as a tool for backcountry travel planning, does raise several issues as a GIScience tool, particularly regarding data reliability and potential biases, such as geographic clustering, subjective terminology and social biases (Flanagin & Metzger, 2008; Parker et al., 2012).

This paper seeks to evaluate the trustworthiness and accuracy of VGI based reports and forecasts. The primary research questions guiding this investigation are:

1. How does the reliability of VGI compare to professionally generated forecasts?

2. What biases and limitations exist in VGI reporting that may affect backcountry users' decision making?

While VGI may be able to improve traditional avalanche forecasting through its more localized and timely data about avalanche and backcountry conditions, uncertainties surrounding its reliability and consistency raise questions about its effectiveness. These concerns are reflected in the larger field of GIScience, as many of these issues are also seen in the broader community discussion regarding the integration of VGI reports and data with traditional methods.

Methodological Framework

This study uses a multi method approach to evaluate the reliability and trustworthiness of Volunteered Geographic Information(VGI) Avalanche reports and forecasts in comparison to professionally generated avalanche forecasts. Rather than engaging in extensive statistical analysis, which would be complex and potentially unproductive given the subjective and inconsistent nature of VGI avalanche reports, this methodology focuses on a systematic literature review and thematic analysis. The approach is primarily theoretical, emphasizing analysis of existing research.

Literature Search Strategy

A systematic search for relevant literature was conducted using web resources and assistance from the university librarian, incorporating evaluation criteria from Flanagin & Metzger (2008) to assess data credibility. A detailed log of the search process, including the process of refining search terms and selecting papers, as shown in **Figure 1**.

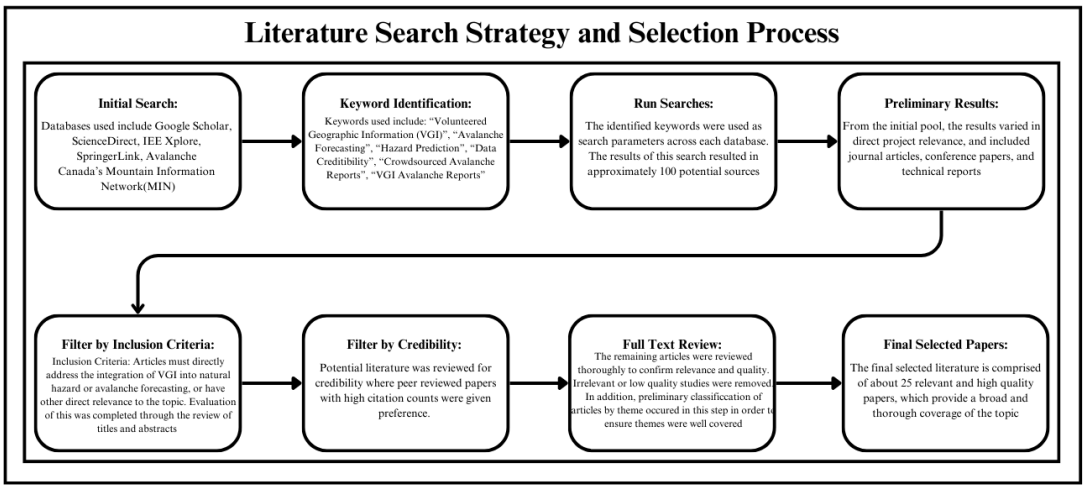


Figure 1: Literature Search Strategy and Selection Process. This flowchart shows the systematic approach to identifying and selecting relevant literature.

Databases and Sources

The literature search used multiple databases, including Google Scholar, ScienceDirect, IEEE Xplore, and SpringerLink. In addition, government reports and data from Avalanche Canada's Mountain Information Network (MIN) were also used. The MIN is a crowd sourced platform established by Avalanche Canada that allows backcountry users to submit observations about snow conditions, weather and avalanche activity.

Search Terms

Keywords: "Volunteered Geographic Information", "VGI", "avalanche forecasting," "hazard prediction", "data credibility", and "VGI avalanche reports" were used. (For a complete list of search terms, please refer to **Appendix A.**)

Inclusion and Organization

Papers were selected based on their relevance to the application or integration of VGI in natural hazard forecasting, evaluating VGI credibility, or direct focus on avalanche forecasting. More emphasis was given to peer reviewed articles with a thorough methodology and high citation count. The selected literary works were then organized into categories based on which theme they best supported, geographic biases, data accuracy, social biases, and subjective risk terminology. **Figure 2** presents a summary of key studies informing this research, including their authors, publication years, key findings and themes they address. This table serves as a reference point for the thematic analysis that follows. A complete table of all key studies is available in **Appendix B**.

Key Studies in VGI Avalanche Reporting Research

Title	Authors/Year	Key Findings	Themes Addressed
The Credibility of Volunteered Geographic Information	Flanagin & Metzger (2008)	VGI credibility depends on source transparency and validation processes They established a criteria for evaluating VG,I which has been used in many other studies	Data credibility and reliability
"Citizen Science in the Age of Neogeography: Utilizing Volunteered Geographic Information for Environmental Monitoring"	Connors, John Patrick, Shufei Lei, and Maggi Kelly (2011)	Citizen science through VGI can improve environmental monitoring, however, consistency can be challenging Citizen science has a democratizing effect, letting a broader range of contributors	Use of VGI, Integration of VGI with traditional methods
"The Role of VGI and PGI in Supporting Outdoor Activities"	Parker, Christopher J., Andrew May, and Val Mitchell (2012)	VGI supports outdoor planning and risk assessment, but should be integrated with traditional data Participatory geographic information can be useful in bridging the gap between community generated data and official datasets.	Integration of VGI

Figure 2: Key studies identified in VGI avalanche reporting research with identified key findings and themes. These key studies were used to inform thematic analysis and provide the foundation for the research which drove this study. This Table is continued in **Appendix B.**

Methodological Limitations

This study has several methodological limitations that should be acknowledged. Firstly, it is largely dependent on the quality and scope of the literature published and selected for review, as it is heavily reliant on literature synthesis. Additionally, this report is primarily focused on Canada, with many of the sources including the use of data and case studies from Avalanche Canada and almost all of the literary sources coming from North America, meaning that the results found may not be representative of all VGI avalanche reporting systems globally. The subjective nature of comparing different studies with varying methodologies also makes it more challenging to form concrete conclusions about VGI reliability when there are conflicting opinions. Finally, as with most technological fields, VGI and VGI reporting platforms are constantly changing, and although attempts were made to select recent studies, findings from earlier studies may not fully apply to the most up to date systems.

Despite these limitations, however, the systematic approach to literature review and thematic analysis provides a solid picture and evaluation regarding the current state of knowledge regarding VGI reliability in avalanche reporting and potential use in forecasting.

Literature Synthesis

Overview of the Field

Avalanche Forecasting systems and methods have changed significantly over time, and with the rise of technology have seen an even more rapid evolution in the past 2-3 decades, with a significant amount of analysis being done through computers and technology. Through this there have been improvements in the availability of this information and data to a wider audience, however, for the most part outside of personal knowledge many people still rely on professional forecasts

like those done by Avalanche Canada. Professional forecasting systems, for example, those used by Avalanche Canada, are heavily reliant on expert analytics of snowpack conditions, weather data, and historical records(Harvey, 2002; Techel et al., 2015). These techniques have proven to be very useful, however, they are limited in both their geographic specificity and real-time responsiveness, as they cover extensive areas with variable weather conditions(Barbolini et al., 2009). To address these limitations VGI reporting was introduced as a form of general hazard prediction, which was quickly adopted into the field of avalanche reporting and forecasting. Researchers such as Connors et al. (2011) and Kocaman and Gokceoglu(2018) have highlighted the potential to use VGI as a tool in combination with traditional methods due to its ability to provide more timely, localized information about avalanche conditions. However, there is debate throughout both the field of avalanche forecasting and the larger field of GIScience regarding the data quality, validation process, and potential intellectual issues associated with the use or reliance on VGI reports and data as a legitimate data source(Flanagin & Metzger, 2008; Parker et al., 2012).

Thematic Grouping

Through the investigation of the literature, several interconnected key themes were found surrounding the use of VGI in Avalanche forecasting. These themes and the primary papers supporting each of them are shown in **Figure 3**. These themes have been identified in order to help to better understand both the strengths and limitations associated with the use of VGI reporting in this context, many of which are also reflected in the wider field of GIScience.

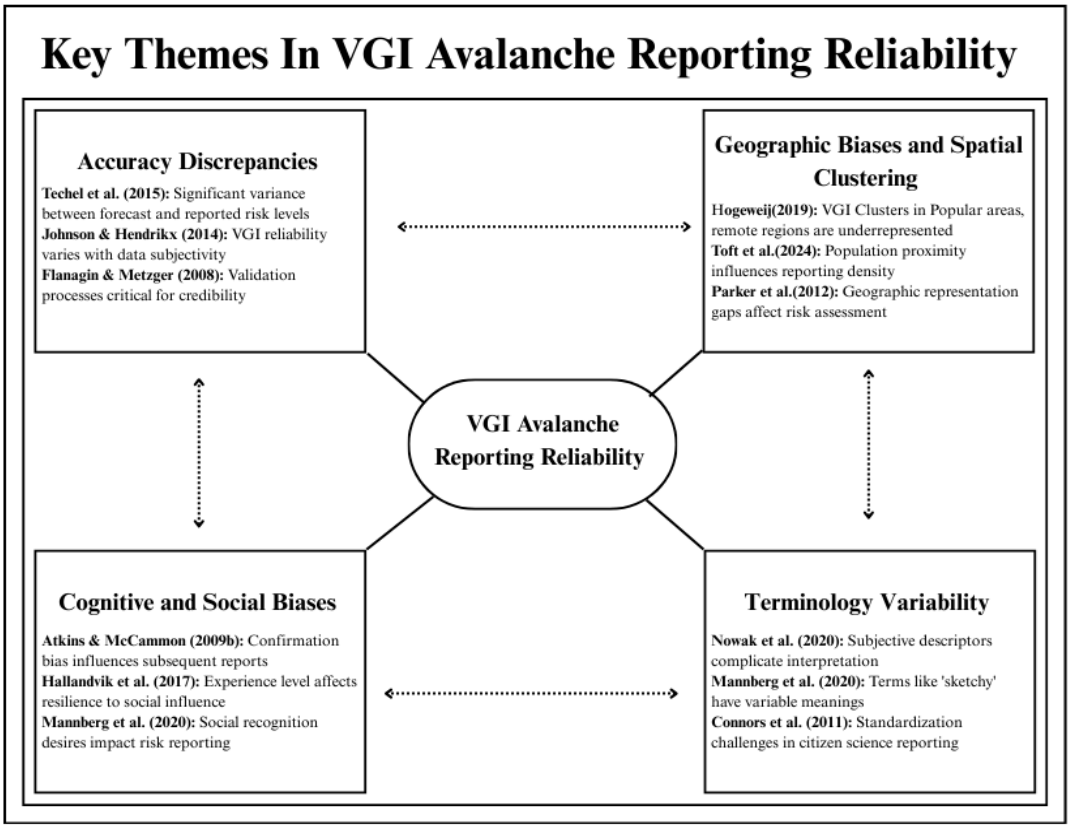


Figure 3: Thematic Framework of VGI Avalanche Reporting Research. The diagram illustrates the 4 key themes identified in the literature and their interconnections

Geographic Biases and Spatial Clustering: A well established issue with the use of VGI is an uneven spread of reports across a region. In Hogeweij(2019) and Toft et al(2024) studies, they found that VGI reports tended to be

more concentrated in more popular backcountry areas or backcountry areas close to areas of high population, whereas in more remote, less popular regions, there was a significant lack of reports. This clustering can lead to gaps in data, which can potentially misrepresent risk in a region or, at the very least, decrease the tools or quality of tools those in the more remote regions have access to. This can lead to those in more remote less travelled regions being more prone to risks associated with avalanches.

Accuracy Discrepancies between Professional Forecasts and VGI: Several studies (e.g., Techel et al., 2015; Flanagin & Metzger, 2008) highlight discrepancies between the risk levels reported by VGI and those provided by professional forecasts. These differences in accuracy can be caused by varying degrees of knowledge among contributors and the subjective nature of VGI reporting. For example, Johnson and Hendrikx found in their 2014 report that while VGI can offer localized updates, its reliability varies significantly in comparison to professionally generated forecasts, highlighting the need for intensive validation systems to ensure reliable data.

Influence of Confirmation Bias and Social Factors: Social influence and cognitive biases are both well known in the field of avalanche safety as hazards, and these same issues are reflected in users' reporting of avalanche conditions. Research by Atkins(2009b) and Hallandvik et al(2017) shows that users may rely on existing reports or be swayed by the consensus of previous reports, leading to confirmation bias in their reports. This bias can also reinforce pre existing perceptions of an area's risk based on past experience, which can also create inaccurate reports or lead to dangerous action. Understanding the impact that social factors can have on the accuracy of VGI is important to understanding whether or not VGI is a legitimate tool to enhance backcountry users' safety.

Variability in Risk Terminology and Subjectivity: One of the challenges identified in the literature is the inconsistent language used in VGI reports. Studies done by Mannberg et al. (2020) and Nowak et al. (2020) explain that subjective descriptors like "seemed safe" or "sketchy" complicate users' ability to assess risk in a region due to their subjective nature. This variability in terminology impacts the effectiveness of VGI data and can make it more complex to integrate into professionally generated forecasting. As a result of this, it may be necessary for a more standard reporting system to be implemented to ensure VGI reporting reliability.

Synthesis and Future Directions:

The reviewed literature demonstrates both the potential benefits and problems associated with integrating VGI into Avalanche forecasting. Due to its more localized nature and its ability to provide more real time data, VGI can offer advantages that traditional avalanche forecasting may not be able to, however, it has challenges associated with its geographic and social biases, discrepancies in accuracy and subjective language.

Systematic Evaluation of VGI Reliability

This section examines the key themes which have emerged from the literature on VGI-based avalanche reporting and argues that although VGI reports provide more localized, real time reporting, its limitations and inconsistencies lead to it being a potential safety concern.

Geographic Clustering and Spatial Bias

A consistent pattern is developed throughout the literature showing that VGI reports tend to be more concentrated in areas where there is higher traffic, which leads to more remote regions being underrepresented. For example, Hogeweij(2019) and Toft et al.(2024) found that in popular areas, there was a much higher number of submissions from users, whereas less popular regions suffered from a distinct lack of submissions. To illustrate this phenomenon, a density-based spatial clustering analysis was performed on 107 VGI avalanche reports submitted, see **Figure 4**. Data was collected from reports to Avalanche Canada's Mountain Information Network(MIN) given between 03/20/25-03/26/25

across Western Canada (British Columbia, Alberta, and the Yukon). The methodological approach for spatial clustering analysis used the density based spatial clustering algorithm implemented in ArcGIS Pro. Clusters were defined as having 4 or more reports that are separated by no more than 30km. The geographic coordinates from each report were collected directly from MIN submissions, and to ensure a fair and unbiased representation of the data, any coordinates which were inconsistent were removed. The analysis revealed that 62.62% of reports fell within clusters, and 37.38% were separate from the clusters. These findings confirm the geographic bias found in the literature, as most of the clusters are found in more popular areas for backcountry recreation. The distribution of the 37.38% of reports outside of the clusters shows the spread of backcountry users, further supporting the point presented by Hogeweij(2019) and Toft et al.(2024) that there are many users who may not have access to adequate information to inform their decisions about backcountry travel and may be affected by the information gap that this clustering creates .

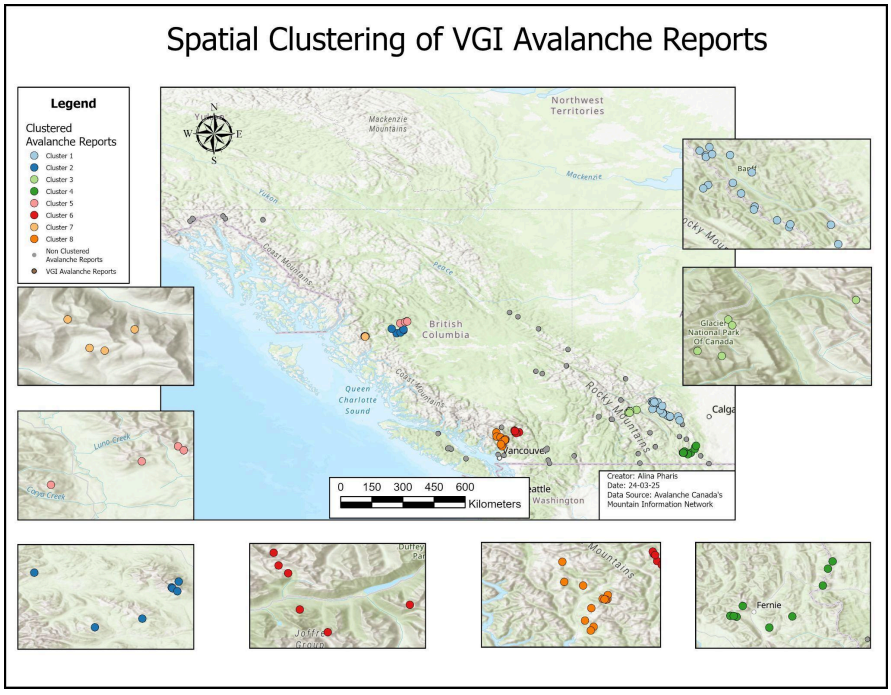


Figure 4: Spatial clustering analysis of 107 VGI avalanche reports from Western Canada between 03/20/25-03/26/25. Colors represent different clusters. 62.62% of reports fall within identifiable clusters.

Accuracy Discrepancies Between VGI and Professional Forecasts

Several studies show noticeable differences between the risk level reported by professional forecasts and what was reported in VGI reports. Although VGI can be seen as a localised, real time alternative to professionally generated forecasts, the discrepancies between the two raises an issue as although it is possible that the discrepancy is occuring due to the geographic specificity or difference in the time or report, it is also possible that it may also be caused by inaccuracies or issues in the VGI data.. John and Hendrikx(2014) found that these discrepancies are often inherent in VGI (crowdsourced) data ,specifically when the nature of the data is more subjective as is the case with VGI avalanche reports. These differences suggest that the way that VGI is currently being used in avalanche reporting may not be as accurate as some may assume, which can make it more challenging to integrate with professional avalanche forecasting. Therefore, many of the papers reviewed suggested that if VGI reports are used they should be used as a supplementary data source as opposed to a main one.

Influence of Cognitive and Social Biases

Research on decision making in avalanche terrain has shown that cognitive biases have a significant impact on risk analysis, which is reflected in VGI reporting. Atkins & McCammon(2009b) and Hallandvik et al.(2017) discuss how confirmation bias and social influence have a significant impact on users' reports, showing that they are often skewed by opinions and consensus from previous reports. For example, once a particular risk level is established by earlier reports, following reports may also be swayed by previous reports, even when the conditions may have changed. This bias impacts the reliability of the data but also can lead to an unsafe narrative being shown to those using VGI reports to inform their backcountry travel plans. This is amplified when looking specifically at less experienced users, as although experienced users are more aware of this bias (as this same bias is a fundamental risk recognized in avalanche safety education as well).therefore, their reports tend to be more accurate to the actual observed conditions, less experienced users may be less confident in their analysis, leading them to an overreliance on the prevailing consensus of previous reports, amplifying the problem.

Variability in Risk Terminology and Subjectivity

One of the most challenging aspects of VGI is that the language used in reports is often subjective. Mannberg et al. 2020 and Nowak et al. (2020) illustrate that descriptors such as “seemed safe” or “sketchy” have a variety of interpretations between users, which makes it harder to determine what the actual reported risk is . The variability in language makes it much more challenging to standardize risk levels between reports and makes it more complicated to integrate VGI reports into forecasting. The challenge stems from a lack of a common reporting framework, which leads to inconsistent data quality and makes it more challenging to use VGI as a tool. Without this standardization, even regions with a large number of VGI reports may not have enough consistent data to be used as a tool.

Argument Development

Through analysis of the literature guided by the themes, an argument about the reliability and trustworthiness of VGI Avalanche reporting and forecasting is established; Although VGI has the potential to improve avalanche forecasting by providing more localized and real time reporting, its should not be relied on as a primary source due to geographic biases, accuracy discrepancies, cognitive and social bias, and subjective reporting. These issues are not exclusive to avalanche reporting and extend into the wider body of GIscience’s users of VGI reporting, but they have implications in both fields, as unless addressed, make VGI data a less useful tool than it could potentially be.

While some argue that the addition of VGI data as a supplemental tool improves the information presented in forecasts (Parker et al., 2012; Connors et al., 2011), there is enough evidence that without a strong validation process and some standardization, the data has too high of a risk of being inaccurate or biased to outweigh these benefits. Expert forecasts may not be geographically specific and have less frequent updates in comparison, but do, however, provide a level of reliability that is currently unmatched by VGI. Therefore, a potentially more effective approach would be a model which can have both the geographic specificity and real time data of VGI with the reliability of professionally generated forecasts. Although this is not currently available, this model would not only mitigate the limitations associated with both the VGI reports and the Professional forecasts but also enhance overall risk assessment and assist backcountry users in their decision making more effectively.

Integration with Practical Examples and Proposed Solutions

To illustrate these points, analysis of MIN data reveals patterns of inconsistencies between VGI avalanche reports and professional forecasts. This aligns with what Nowak et al. (2020) describe as the ‘ambiguity’ in avalanche forecasting, where differences in reporting can cause variability in the interpretation of data. For example, VGI reports for popular backcountry areas report risk differently from professionally generated forecasts. Based on previous research, these discrepancies likely stem from some combination of geographic clustering and confirmation bias.

Improved Reporting Platforms: Features like confidence scores and an expert verification process provide a potential option to improve the credibility of VGI reports. For example, a proposed validation workflow (see

Figure 5*)* outlines a process where a user submits an avalanche report, which then undergoes several screening processes to ensure that the data being shown to users can be confirmed as accurate to a reasonable degree before it is assigned a confidence score and is displayed for users. This workflow is designed to enhance data reliability, similar to the citizen science strategies suggested by Arienzo et al. (2021), integrating structured feedback and training for VGI contributors could enhance the overall data quality."

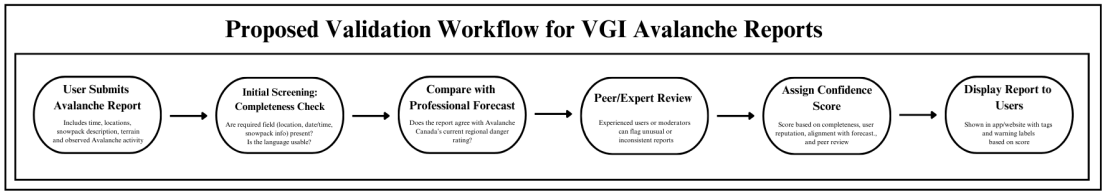


Figure 5: Proposed validation workflow to improve VGI avalanche report reliability

Data Integration: Combining information from VGI reports with professional forecasts can provide a more holistic picture of backcountry conditions by using the strengths of both systems.

Standardizing Reporting Protocols: Implementing clearer guidelines for risk terminology would allow for users to better understand the subjective language and ensure more continuity between reports.

User Training: Providing more accessible options for backcountry users to educate themselves on accurate reporting techniques, as well as reading and interpreting both VGI reports and professionally generated forecasts, can help users more effectively use both tools, both as a viewer and submitter.

Conclusion of Analysis and Argumentation

In summary, the critical analysis shows that while VGI introduces valuable localized real time data about avalanche/backcountry conditions, it faces challenges associated with its reliability and accuracy; Geographic clustering, discrepancies with professional forecasts, cognitive and social biases, and the variability of risk terminology. In order to address many of these challenges, the integration of VGI (with improved reporting methods) reports with professional forecasts given that any VGI data included has undergone a screening process in order to ensure data reliability like the one seen in **Figure 5**, would help to make backcountry users safer and improve the reliability and trustworthiness of VGI reports.

Discussion and Synthesis

VGI reporting is a promising new technology providing localized, more realtime avalanche data, however, it continues to face several challenges that limit its use as an independent forecasting/reporting tool. Those issues include spatial clustering (more reports in more popular areas, remote areas are underrepresented) (Hogeweij, 2019; Toft et al., 2024), discrepancies between VGI and professional forecasting(Techel et al., 2015; Johnson & Hendrikx, 2014), variable risk terminology (Mannberg et al., 2020; Nowak et al., 2020), and cognitive bias which is both an independent issue(McCammon, 2009b; Hallandvik et al., 2017).

One of the biggest challenges identified in the literature that was faced was the reliance on self-reported data, as it introduced biases which were very difficult to mitigate without additional information. In addition, the variability in data quality from the VGI made it more challenging to compare directly with professionally generated avalanche forecasts, which further highlighted the value of implementing a validation system and a standardized reporting framework, as highlighted by Flanagin & Metzger (2008). These challenges are not exclusive to the use of VGI in avalanche forecasting but are seen across GIScience, and finding a solution to address them would benefit numerous sectors.

The issues identified in this study are reflective of wider concerns and issues within GIScience regarding the reliability of VGI data. The challenges of geographic clustering and data quality variability seen in avalanche reporting are mirrored in other VGI applications such as OpenStreetMap and citizen science mapping projects. Goodchild and Li(2012) have similarly emphasized that reliable VGI requires robust validation systems.. These findings align with what this study found about avalanche reporting. The spatial biases observed in VGI avalanche reporting reflect what Haklay(2013) calls the "participation inequality" problem in crowdsourced geographic data; the distribution of data creates uneven data landscapes. These connections show that issues in VGI avalanche reporting and VGI data reliability are part of broader GIScience challenges surrounding data accessibility, timely reporting and maintaining data quality standards.

While this paper argues that if VGI data from avalanche reports is more validated, standardized, and integrated in professional forecasts, it will be more useful, some researchers present opposing viewpoints. Parker et al.(2012) suggest that excessive standardization may reduce VGI's ability to represent the local conditions and reduce the value of experiential knowledge which exists in reports without the standardization. Atkins & McCammon (2004) support this statement as they found that there is significant benefit in experiential knowledge within the field of professional avalanche forecasting and snow science, therefore it would follow that this same experiential knowledge may also be useful in VGI reports.Some backcountry users and researchers also say that imperfect real time information provides value in rapidly changing mountain conditions where professional forecasts may be insufficient or outdated. Johnson & Hendrikx(2014) suggest that the geographic specificity and real time nature of VGI reporting may be more important than concerns about reliability in certain contexts.

These alternative perspectives highlight the complex balance between ensuring data reliability and ensuring that the benefits that VGI avalanche reporting offers are not lost. Integrating these discoveries with existing research suggests that an approach to avalanche reporting and forecasting which combines VGI reporting with professionally generated forecasts, would provide the most effective forecasting. This integrated strategy aligns with the perspectives in the papers reviewed by Parker et al.(2012) and Connors et al. (2011), which emphasize the importance and value of a diverse data set when making decisions in the backcountry.

Future research should focus on improving the validation and standardization of VGI data, finding methods which don't significantly impact the benefits of VGI, the integration of machine learning technologies and AI, and finding a way to effectively integrate VGI report data with professional avalanche forecasts. These efforts are important as, if effectively implemented, they will allow for VGI to become a more useful and legitimate tool for avalanche safety and backcountry decision making.

Conclusion

This study has evaluated the reliability and trustworthiness of VGI in avalanche forecasting through the examination of the impact of geographic clustering, accuracy discrepancies, cognitive and social biases, and subjective reporting.

This study provides several key insights about the initial research questions which were used to guide this study. Regarding the comparison of the reliability of VGI report data and professionally generated forecasts, it was found that although VGI reports do offer geographic specificity and real time updates which are lacking from professional forecasts, it is in many cases much less reliable due to inconsistencies in reporting and variability in data quality. Concerning potential biases and limitation of VGI data, several were identified. VGI avalanche reports are prone to geographic clustering, confirmation (cognitive and social) bias, and variability in risk and descriptive terminology used in reports. All of these limitations and biases have the potential to impact VGI avalanche reports to be used as a trustworthy and reliable source for backcountry users.

It was concluded that while VGI provides a data source that can deliver real time localized data, as an independent source it does not have the reliability and trustworthiness to be reliable, however, if integrated into a professionally generated forecast it can serve as a useful supplementary tool provided necessary validation and standardization steps were taken. These steps for validation and standardization may take several forms, including the implementation of expert

validation, the development of standardized reporting guidelines, and the integration of machine learning or AI technologies to improve these reporting platforms. With this the aim is to enhance the overall accuracy and usability of VGI data, ideally leading to improved safety by increasing the tools which backcountry users have access to to make informed decisions about their travel in backcountry terrain.

These findings contribute to ongoing debates in GIScience concerning the use of VGI data and help contribute to directing this research. Continued exploration in this area is important to developing more reliable and useful forecasting, which can help increase backcountry users' safety.

References:

Arienzo, M. M., Collins, M., & Jennings, K. S. (2021). Enhancing engagement of citizen scientists to monitor precipitation phase. *Frontiers in Earth Science, 9*(February). https://doi.org/10.3389/feart.2021.617594

Atkins, D., & McCammon, I. (2004). Differences between avalanche experts and novices. In *Proceedings of the 2004 International Snow Science Workshop* (pp. 452–461). Retrieved from http://arc.lib.montana.edu/snow-science/objects/issw-2004-452-461.pdf

Atkins, D., & McCammon, I. (2009b). Amateur decision-making in avalanche terrain with and without a decision aid: A stated choice survey. *Natural Hazards, 52*(1), 185–209. https://doi.org/10.1007/s11069-009-9365-4

Barbolini, M., Pagliardi, M., Ferro, F., & Corradeghini, P. (2009). Avalanche hazard mapping over large undocumented areas. *Natural Hazards, 56*(2), 451–464. https://doi.org/10.1007/s11069-009-9434-8

BC Coroners Service. "Winter Activity Deaths in British Columbia 2008/2009 – 2017/2018," n.d. https://www2.gov.bc.ca/assets/gov/birth-adoption-death-marriage-and-divorce/deaths/coroners-service/statistical/winter-activity.pdf.

Connors, J. P., Lei, S., & Kelly, M. (2011). Citizen science in the age of neogeography: Utilizing volunteered geographic information for environmental monitoring. *Annals of the Association of American Geographers, 102*(6), 1267–1289. https://doi.org/10.1080/00045608.2011.627058

Flanagin, A. J., & Metzger, M. J. (2008). The credibility of volunteered geographic information. *GeoJournal, 72*(3–4), 137–148. https://doi.org/10.1007/s10708-008-9188-y

Goodchild, Michael F. "Citizens as Sensors: The World of Volunteered Geography." GeoJournal 69, no. 4 (November 19, 2007): 211–21. https://doi.org/10.1007/s10708-007-9111-y.

Goodchild, Michael F., and Linna Li. "Assuring the Quality of Volunteered Geographic Information." Spatial Statistics 1 (May 1, 2012): 110–20. https://doi.org/10.1016/j.spasta.2012.03.002.

Hallandvik, L., Andresen, M. S., & Aadland, E. (2017). Decision-making in avalanche terrain – How does assessment of terrain, reading of avalanche forecast and environmental observations differ by skiers' skill level? *Journal of Outdoor Recreation and Tourism, 20*(November), 45–51. https://doi.org/10.1016/j.jort.2017.09.004

Harvey, S. (2002). Avalanche incidents in Switzerland in relation to the predicted danger degree. In *2002 International Snow Science Workshop* (pp. 443–448). Retrieved from https://arc.lib.montana.edu/snow-science/objects/issw-2002-443-448.pdf

Haklay, Muki. "Citizen Science and Volunteered Geographic Information: Overview and Typology of Participation." In Springer eBooks, 105–22, 2012. https://doi.org/10.1007/978-94-007-4587-2_7.

Hogeweij, J. D. M. (2019). Exposure to avalanche terrain in crowd-sourced ski touring route plans. Retrieved from http://dspace.library.uu.nl/handle/1874/394010

Johnson, J., & Hendrikx, J. (2014). Using global crowd-sourced data to understand travel behavior in avalanche terrain. In *International Snow Science Workshop 2014 Proceedings* (pp. 249–253). Retrieved from https://arc.lib.montana.edu/snow-science/objects/ISSW14_paper_O9.04.pdf

Kocaman, S., & Gokceoglu, C. (2018). On the use of CITSCI and VGI in natural hazard assessment. In *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences XLII–5* (pp. 69–73). https://doi.org/10.5194/isprs-archives-xlii-5-69-2018

Mannberg, A., Hendrikx, J., & Johnson, J. (2020). Risky positioning – Social aspirations and risk-taking behaviour in avalanche terrain. *Leisure Studies, 40*(4), 495–512. https://doi.org/10.1080/02614367.2020.1831046

Nowak, S., Bartram, L., & Haegeli, P. (2020). Designing for ambiguity: Visual analytics in avalanche forecasting. In *IEEE* (pp. 81–85). https://doi.org/10.1109/vis47514.2020.00023

Parker, C. J., May, A., & Mitchell, V. (2012). The role of VGI and PGI in supporting outdoor activities. *Applied Ergonomics, 44*(6), 886–894. https://doi.org/10.1016/j.apergo.2012.04.013

Techel, F., Zweifel, B., & Winkler, K. (2015). Analysis of avalanche risk factors in backcountry terrain based on usage frequency and accident data in Switzerland. *Natural Hazards and Earth System Sciences, 15*(9), 1985–1997. https://doi.org/10.5194/nhess-15-1985-2015

Toft, H. B., Karlsen, K., Landrø, M., Mannberg, A., Hendrikx, J., & Hetland, A. (2024). Who skis where, when? – A method to enumerate backcountry usage. *Cold Regions Science and Technology, 104370*. https://doi.org/10.1016/j.coldregions.2024.104370

Appendix A

Primary Terms	Secondary Terms	Tertiary Terms
“Volunteered Geographic Information	"avalanche forecasting"	"reliability"
"VGI"	"avalanche prediction"	"accuracy"
"crowdsourced data"	"Natural hazard assessment"	"credibility"
"user-generated content"	"backcountry safety"	"validation"
"citizen science"	"risk reporting"	"standardization"
"participatory GIS"	"Avalanche reports"	
	"mountain safety"	
	"winter recreation hazards"	

Appendix B

Title	Authors/Year	Key Findings	Themes Addressed
The Credibility of Volunteered Geographic Information	Flanagin & Metzger (2008)	VGI credibility depends on source transparency and validation processes They established a criteria for evaluating VGI which has been used in many other studies	Data credibility and reliability
"Citizen Science in the Age of Neogeography: Utilizing Volunteered Geographic Information for Environmental Monitoring"	Connors, John Patrick, Shufei Lei, and Maggi Kelly (2011)	Citizen science through VGI can improve environmental monitoring, however, consistency can be challenging Citizen science has a democratizing effect, letting a broader range of contributors	Use of VGI, Integration of VGI with traditional methods
"On the Use of CITSCI and VGI in Natural Hazard Assessment"	Kocaman & Gokceoglu (2018)	VGI shows promise in natural hazard assessment, but requires improved validation and standardization Citizen science can significantly improve spatial resolution of hazard maps, and can fill in gaps left by conventional methods	Integration of VGI in hazard forecasting
“Analysis of Avalanche Risk Factors in Backcountry Terrain Based on Usage Frequency and Accident Data in Switzerland"	Techel, F., B. Zweifel, and K. Winkler (2015)	Significant discrepancies exist between reported avalanche risk and actual incidents There is a direct correlation between backcountry usage and avalanche accident rates, showing the relation between human behaviour and hazard exposure.	Accuracy discrepancies, risk assessment
"Using Global Crowd-Sourced Data to Understand Travel Behavior in Avalanche Terrain"	Johnson & Hendrikx (2014)	Variability in risk assessment and subjective reporting are evident in VGI data Crowd sourced data captures local details, however, there observations often vary widely, making comparison more complicated	Subjectivity, reliably of data
"Risky Positioning – Social Aspirations and Risk-taking Behaviour in Avalanche Terrain"	Mannberg, Andrea, Jordy Hendrikx, and Jerry Johnson (2020)	Desire for social recognition among contributors causes risky behaviour, which may impact reported risk.	Subjectivity of VGI reports, data standardization
"Designing for Ambiguity: Visual Analytics in Avalanche Forecasting"	Nowak, Stan, Lyn Bartram, and Pascal Haegeli (2020)	Proposed methods to visually represent data ambiguity, helping to interpret inconsistent or uncertain data Enhancing visual analytics tools can reduce challenges of interpreting inconsistent data.	Visualization, data ambiguity
"Exposure to Avalanche Terrain in Crowd-sourced Ski Touring Route Plans"	Hogeweij (2019)	Demonstrated that VGI reports are clustered in popular areas, leaving remote regions under represented Lack of reports in less frequented areas may lead to underestimation of risk	Geographic clustering, spatial bias
"The Role of VGI and PGI in Supporting Outdoor Activities"	Parker, Christopher J., Andrew May, and Val Mitchell (2012)	VGI supports outdoor planning and risk assessment but should be integrated with traditional data Participatory geographic information can be useful in bridging the gap between community generated data and official datasets.	Integration of VGI